

Experiment 2

Module Name: Robot Framework

Experiment Name: Variables and Data-Driven Automation

Problem Statement: Implement a test case that uses variables to store data like usernames and passwords. Create a data-driven test that reads test data from an external file and performs the same set of actions on multiple data sets.

Objective: To separate test data from test logic using variable dictionaries and keyword templates for Data-Driven Testing (DDT).

Tools, Software, and Concepts Used:

Robot Framework Test Templates, Variable sections (`*** Variables ***`), Keyword arguments (`[Arguments]`)

Source Code / Implementation Steps:

1. Create a folder inside that folder make a file 'Experiment2.robot' and write this:

```
*** Settings ***
Library      SeleniumLibrary
Test Template  Invalid Login Scenario

*** Variables ***
${URL}        https://www.saucedemo.com/
${BROWSER}    Chrome

*** Test Cases ***
Invalid User 1      USERNAME      PASSWORD
                    locked_out_user secret_sauce
Invalid User 2      invalid_user  wrong_pass
Empty Credentials   ${EMPTY}      ${EMPTY}

*** Keywords ***
Invalid Login Scenario
    [Arguments]    ${username}    ${password}
    Open Browser    ${URL}        ${BROWSER}
    Input Text      id=user-name    ${username}
    Input Text      id=password     ${password}
    Click Button    id=login-button
    Page Should Contain Element    css=.error-message-
container
    Close Browser
```

2. From the folder go to cmd and write this

```
>>> python -m robot Experiment2.robot
```

Output with clearly labelled screenshots:

```
D:\My Desktop\Robot Project>python -m robot Experiment2.robot
=====
Experiment2
=====
Invalid User 1                                     | PASS |
-----
Invalid User 2                                     | PASS |
-----
Empty Credentials                                 | FAIL |
Element with locator 'id=user-name' not found.
-----
Experiment2                                     | FAIL |
3 tests, 2 passed, 1 failed
=====
Output: D:\My Desktop\Robot Project\output.xml
Log:    D:\My Desktop\Robot Project\log.html
Report: D:\My Desktop\Robot Project\report.html
```

Screenshot 1: Terminal Output

Brief Result: The single test template executed three times sequentially, injecting different username/password combinations each time.

Observation: The Test Template setting overrides the default test case behavior, turning the test cases section into a localized data table.

Conclusion: DDT in Robot Framework significantly reduces code duplication and makes expanding test coverage as simple as adding a new row of data.